Revision log for "On quadratic residue codes and hyperelliptic curves"

David Joyner Claude (Anthropic)

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Scope

Original: `original/qrcodes-hyperell-dmtcs2.tex`, the version submitted to DMTCS (received 20 October 2006), sole author D. Joyner.
Revised: `qrcodes-hyperelliptic.tex`, D. Joyner and Claude.

The published 2008 DMTCS version (doi:10.46298/dmtcs.429, arXiv:math/0609562v2) is a *different* revision of the same submission. This log compares only against the 2006 file. In the original, numbering ran with one counter per environment; in the revision, numbering is by section (for example, Theorem 4.2). The table in Section 6 maps old results to new ones.

Every numerical claim in either version was checked with the Python code in `code/`, using exhaustive enumeration of C_{NQ} for $p \leq 23$.

1 Format and front matter

1. Document class changed from `dmtcs` to `article`, 12pt, letter paper, Times (`mathptmx`), compiled with `pdflatex`. The RCS macros, `latin1` input encoding, `subfigure` and `natbib` were removed.

2. Author block: the affiliation "Mathematics Dept., USNA, Annapolis, MD" was replaced by `wdjoyner@gmail.com`. Claude (Anthropic) was added as second author, with a footnote and an acknowledgement paragraph saying what it contributed.

3. The abstract was rewritten to state the corrected results and constants. Keywords were updated, and the MSC codes were updated to MSC 2020.

4. A table of contents was added.

5. The material after `\end{document}` (an unused remark with an unproved lower bound on $|S_3|$) was deleted.

2 Mathematical corrections

2.1 Errors corrected

1. **Lemma 1 (old), parity claims.** We have $r_{S_1}^* r_{S_2}^* = r_{S_1} r_{S_2} + (|S_1| + |S_2| + 1)J$, and $(r_{S_1} r_{S_2}) \equiv |S_1||S_2|$. So in the cases (even, even) and (odd, odd) the element $(r_{S_1} r_{S_2})^*$ has odd and even weight respectively, the opposite of what was stated. The lemma was replaced by the short list of facts in §2.1.

2. **"$*$ is an algebra involution on R_{even}."** This is false: $*$ is not additive, since $(a+b)^* = a+b+J$, and it maps even-weight elements to odd-weight ones. Also, $R_{even} = (1+x)R$ contains no 1. The claim was deleted.

3. **Remark 1 (old).** "C_{NQ} contains a QR code" was corrected to "$C_{NQ} \subseteq \mathrm{QR} \oplus \mathrm{QR}'$" (Remark 3.12).

4. **Lemma 2 (old)** relied on self-duality, which was stated only as a conjecture (Conjecture 3, old). Self-duality is now proved (Theorem 3.4), so the lemma is justified. The bound is now attributed correctly to Rains [2], not Sloane–Mallows.

5. **Corollary 1 (old),** $\max \#X_S > \frac{5}{3}p - 4$**.** The proof assumed that a minimum-weight codeword comes from an even-size S. For $p = 11$ it does not: the 77 weight-6 words all come from odd S. The result was restated and reproved as Corollary 4.5, $M_p \geq \frac{5}{3}p - 6$ for $p \equiv 3 \pmod 4$ and $p \geq 11$, using the new twist identity.

6. **Remark after Proposition 1 (old).** The text gave both "$\sqrt{p}+1$" and "$\sqrt{p}+2$", and applied part (c) to the wrong parity. The corrected statement is Remark 3.12. That remark also notes that the bound follows from Weil's bound and the trivial count. (The referee report said this bound was stronger than Weil; that was wrong and is corrected here.)

7. **Theorem (old 2), proof.** The entropy function had a spurious leading δ. The weight formula now cites Proposition 1, not Lemma 3.

8. **Theorem (old 3), proof.** The proof claimed to contradict "$B(1.62, p)$ for all sufficiently large p". The correct logic is that $B(1.62, p)$ can hold for only finitely many p. Rewritten as Theorem 4.3 with the sharp constant $c^* = 1.62584\ldots$.

9. **Theorem 4 (old, LQR parameters).** For $p \equiv 3 \pmod 4$, the claimed lower bound $d \geq 4p - 2\max \#X_S$ was not proved. The injectivity argument was incorrect: it asserted $r_Q r_{T^c} = 0$, and it concluded "$S_1 = \emptyset$, $S_2 = \mathbb{F}_p$", contradicting the injectivity it claimed. For $p \equiv 1 \pmod 4$, d_p should be $4p + 4 - 2M$. The statement "this code is non-linear" is false for $p \equiv 1 \pmod 4$.

10. **Remark after Theorem 4 (old).** The remark doubted that the minimum distance could be below $2p$. In fact it always is: the minimum distance is $2d(C_{NQ})$, for example $8 < 14$ at $p = 7$.

11. **Lemma 7 (old).** For $p \equiv 1 \pmod 4$ and odd S, the sum should be over f_{S^c}.

12. **Example (old, LQR, $p = 11$).** The two sets $S = \{1,2,3,4\}$ and $\{1,2,3\}$ were interchanged. The section was removed (see Section 3).

13. **Section 3 (old), genus.** The formula $\gamma = n+k+1-d$ was corrected to $\gamma = n+1-k-d$.

14. **Example (old, $p = 13$ zeta).** The text said "only 8 of the 12 zeros". $P(T)$ has degree 16, and exactly 12 of its 16 zeros lie on $|T| = 1/\sqrt{2}$. The coefficients were correct.

15. **Optimality of the $[26,13,6]$ code.** The claim rested on general linear-code tables, which, as we understand them, allow $d = 7$ for $[26,13]$. It was restated as optimality among even codes, with an explicit caveat that this has not been verified independently.

16. **Voloch's Theorem (old 6).** The constant window $-\frac{1}{2} \le a \le \frac{5}{2}$ holds only for $\ell = 2$. It was replaced by $O(1) \in [-2,3]$, and "$\ell > 2$" versus "$\ell \ge 2$" was made consistent. The introduction's claim that $\#X_Q > 1.5p$ for $p \equiv 3 \pmod 8$ was corrected to $\#X_Q = 1.5p - \frac{1}{2}$.

17. **Conjecture 3 (old).** "$C \cap C^{\perp} = \emptyset$" was corrected to $\{0\}$, and "direct product" to "direct sum".

2.2 Former conjectures now proved

1. Conjecture 3 (old): self-dual for $p \equiv 3 \pmod 4$, LCD for $p \equiv 1 \pmod 4$. Now Theorem 3.4, with a three-line proof via $\langle u, v \rangle = \mathrm{ct}(uv^{-})$.

2. Conjecture 4 (old): C' is formally self-dual with distribution $A + A^{*}$. Now Theorem 6.2: $C'^{\perp}$ is the half-swap of C', so C' is isodual. The distribution statement is immediate.

3. Section 3 (old) asked whether C_{NQ} always satisfies RH for $p \equiv 3 \pmod 4$. Answered negatively: RH fails at $p = 23$ (30 of 32 zeros on the circle). This agrees with Boston–Hao [1].

2.3 New results

1. Twist identity (Lemma 2.4): $\#X_T + \#X_{T'} = 2p+2$ when $\chi(f_T(a)) = -1$, and its consequence $M_p + m_p \ge 2p+2$ (Corollary 2.6).

2. Even subcode C_{ev} (Proposition 3.10). It has dimension $p-1$ for every p, weights $2p + 2 - \#X_T$, and $d(C_{\mathrm{ev}}) = 2p+2-M_p$. For $p \equiv 3 \pmod 4$ it is the doubly-even subcode, so $\#X_T \equiv 0 \pmod 4$ for even T.

3. Theorem 4.1 (Goppa): the congruence condition $p \equiv 1 \pmod 4$ was removed, and the threshold sharpened to $c < 1.7799$. This supersedes the LQR result (old Theorem 5, $c = 1.57$).

4. Theorem 4.3 (MRRW): valid for all p, with $c^{*} = 1.62584$.

5. Theorem 5.1(3): the case $p \equiv 5 \pmod 8$, $S = Q \cup \{0\}$, which gives $\#X_S = 1.5p + \frac{1}{2}$. Every prime $p \ge 11$ now has an explicit S with $\#X_S \ge 1.5p - \frac{1}{2}$.

6. New data: weight distributions for $p = 17, 19, 23$ (Table 1), the RH status of seven codes (Table 2), and M_p, m_p for $p \le 23$ (Table 3).

3 Material removed

1. Section 4 (old), *Long quadratic residue codes*: Lemmas 5–7, Theorem 4, Theorem 5, and the LQR example. They were replaced by Remark 4.7, which explains that the LQR code is a doubled code or a coset of one, and so has the same relative distance as C_{NQ} at half the rate. The even subcode gives the stronger result.

2. The "Cyclotomic arithmetic" section was condensed to the facts actually used. The sets $H(S_1, S_2, a)$ and identity (2) (old) were removed.

3. The quoted lemma "Bazzi–Mitter Prop. 3.3" (old Lemma 4) was absorbed into Proposition 3.6.

4 Additions for readability

1. Eleven worked examples (Examples 2.1, 2.3, 2.5, 3.3, 3.7, 3.8, 3.11, 6.1, 6.3 and two in Sections 4 and 5). They cover arithmetic in R, point counting via Legendre sums, the twist identity, dimension, the weight formula, the even subcode, Voloch's construction for $p = 13$, and zeta polynomials.

2. Proofs were rewritten in full, including the dimension count (Proposition 3.2) and Lemma 2.2 (Artin–Hasse–Weil) with the trivial bounds (3).

3. Section 7, *Open questions and their difficulty*: a random-code heuristic ($M_p \approx 1.78p$) and an assessment of each open question.

4. Section 8, *Literature*, in two parts: before 2006 and 2006 and after.

5. Appendix A, *Glossary*: 54 entries.

6. Typos fixed, among others: McEliese → McEliece, Welsh → Welch, Tsafsman → Tsfasman, Dordrechet → Dordrecht, "This rest of", "the the", "if and only if and only if", "consquence", "we if"; the cross-reference "Theorem \ref{cor:main}"; and the inconsistent "$p > 5$". The second supplementary law is no longer called "quadratic reciprocity".

5 Bibliography

1. Updated: Bazzi–Mitter (IEEE IT 2006), Jiang–Vardy (IEEE IT 2004), Joyner (DMTCS 2008).

2. Added: Alahmadi–Özdemir–Solé 2018, Boston–Hao 2017, Calderbank 1983, Chen–Peterson–Weldon 1969, Chinen 2017/18, Grassl tables, Joyner–Kim 2011, Karlin 1969, Kasami 1974, Mallows–Sloane 1973, Massey 1992, MRRW 1977, Rains 1998, Shi–Xu–Solé 2021.

3. Removed as uncited: SAGE 2.10, Wage 2006, Chinen's "An abundance of invariant polynomials". The "Wage" remark on $|S| = 2, 3$ was dropped.

4. Bibliographic details marked from memory in the referee report should be checked against the publishers' pages before submission.

6 Map of results

2006 version	Revision
Lemma 1 (products)	§2.1 facts (corrected)
Lemma 2 (mindist bound)	used in Corollary 4.5 (Rains)
Lemma 3 (Artin–Hasse–Weil)	Lemma 2.2
Lemma 4 (Bazzi–Mitter)	absorbed into Proposition 3.6
Proposition 1	Proposition 3.6
Corollary 1 ($\frac{5}{3}p-4$)	Corollary 4.5 ($\frac{5}{3}p-6$, new proof)
Theorem 1 (Bazzi–Mitter)	Theorem 4.1 (no congruence condition)
Theorem 2 (1.77, $p \equiv 1 \bmod 4$)	Theorem 4.1 ($c < 1.7799$, all p)
Theorem 3 (1.62, $p \equiv 1 \bmod 4$)	Theorem 4.3 ($c < 1.62584$, all p)
Corollary 2 (1.62, all p)	Theorem 4.3
Conjecture 3 (self-dual / LCD)	Theorem 3.4 (proved)
Conjecture 4 (C')	Theorem 6.2 (proved)
Section 4: Theorem 4, Theorem 5, Lemmas 5–7	Remark 4.7 (removed)
Lemma 8 (Voloch)	Theorem 5.1 (with new case $p \equiv 5 \bmod 8$)
Theorem 6 (Voloch, Chebotarev)	Theorem 5.3
—	Lemma 2.4, Corollary 2.6, Proposition 3.10, Theorem 6.2 (new)

References

[1] N. Boston and J. Hao, The weight distribution of quasi-quadratic residue codes, arXiv:1705.06413.

[2] E. M. Rains, Shadow bounds for self-dual codes, *IEEE Trans. Inform. Theory* 44 (1998), 134–139.